IT203 **Lab 2: Passwd Cracking**

Max points: 10

In this assignment you'll install Kali Linux 2024 in VirtualBox. Then you will create new groups and users based on the table and crack Passwords on a Linux System using John the Ripper John. You'll access the Ubuntu machine to carry out your work.

The following link will show you how to install Kali Linux 2024 in VirtualBox on Ubuntu/Windows PC. The version we will be installing is Kali Linux 2024.

How To Install Kali Linux 2024 in VirtualBox (Windows)
https://www.youtube.com/watch?v=sAMnXte56yY

How to Install Kali inside VMware (Ubuntu)
https://www.kali.org/docs/virtualization/install-vmware-guest-vm/

After installing the Kali Linux, you can keep continuing the rest of the lab assignment.

A. Creating User Accounts and Groups

New groups and users will be created based on the tables below:

Group: seniors	
user	password
elmo	123123
oscar	sanjose

Group: juniors	
user	password
lisa	virginia
homer	acapulco

1) Populate the group list first by adding two more groups: juniors and seniors.

```
groupadd juniors
groupadd seniors
```

2) Confirm that the groups have been added.

```
cat /etc/group | grep "seniors"
```

Notice the group ID 1002.

```
cat /etc/group | grep "juniors"
```

Notice the group ID 1001.

3) Type the command below to view the user accounts on the system.

```
cat /etc/passwd
```

4) Add users elmo and oscar and assign them to the group seniors.

```
useradd elmo -g seniors
useradd oscar -g seniors
```

5) Add users lisa and homer and assign them to the group juniors.

```
useradd lisa -g juniors
useradd homer -g juniors
```

6) View the /etc/passwd file once more to verify that all accounts are successfully created.

```
cat /etc/passwd
```

7) Configure passwords for the users: elmo, oscar, lisa and homer.

```
passwd elmo
```

8) Type 123123 twice as the password.

```
passwd oscar
```

9) Type sanjose twice as the password.

```
passwd lisa
```

10) Type virginia twice as the password.

```
passwd homer
```

11) Type acapulco twice as the password.

12) Type the command below to view the created users in the shadow file.

```
sudo tail -n -4 /etc/shadow
```

Leave the Terminal open for the next task.

B. Cracking Passwords on a Linux System Using John the Ripper

1) Before using John the Ripper, type the command below within a Terminal window to view the available options that can be used with the application.

```
john -help
```

2) Download `john.txt` file from Canvas and save it in your current directory.
3) unshadow the passwd & shadow file and save them as mypasswd

```
sudo unshadow /etc/passwd /etc/shadow >  mypasswd
```

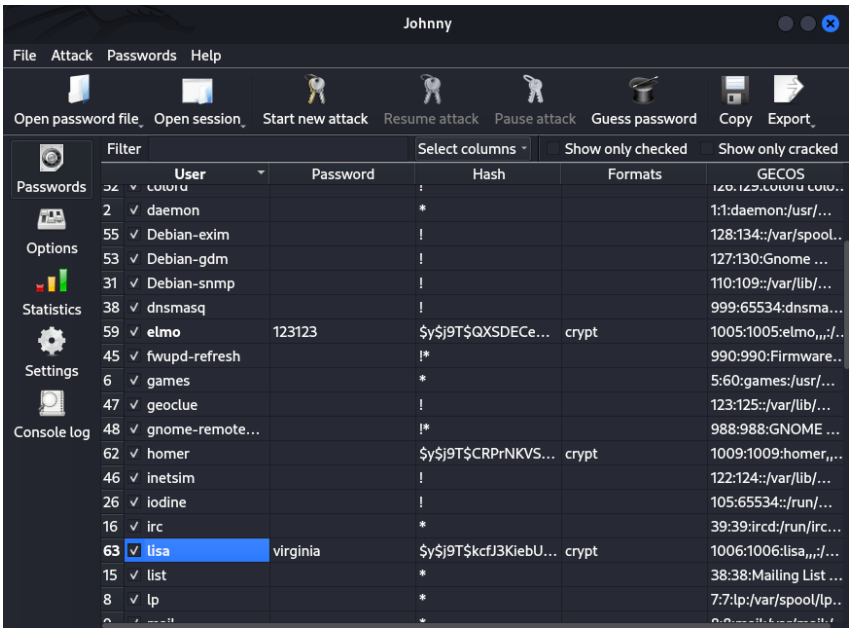
4) Run John the Ripper against the mypasswd file using default crypt format.

```
sudo john --format=crypt -wordlist=/home/checkout/john.txt mypasswd
```

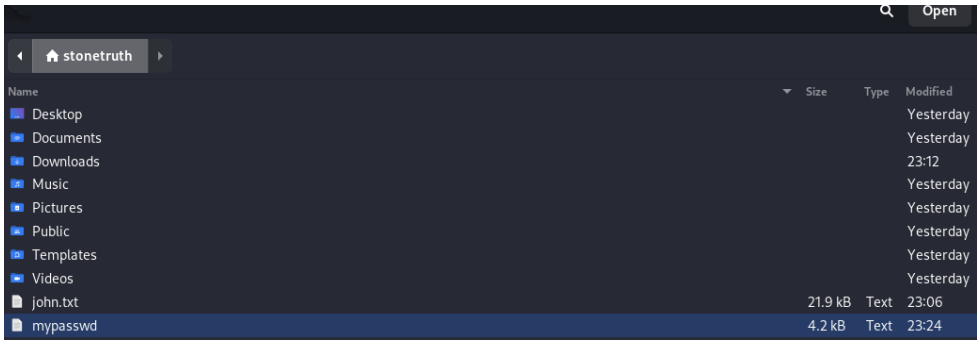
5) See what happened. Did it successfully crack passwords?
   Yes, it successfully cracked the passwords.

6) Type the command below to view johnny GUI.

```
johnny
```

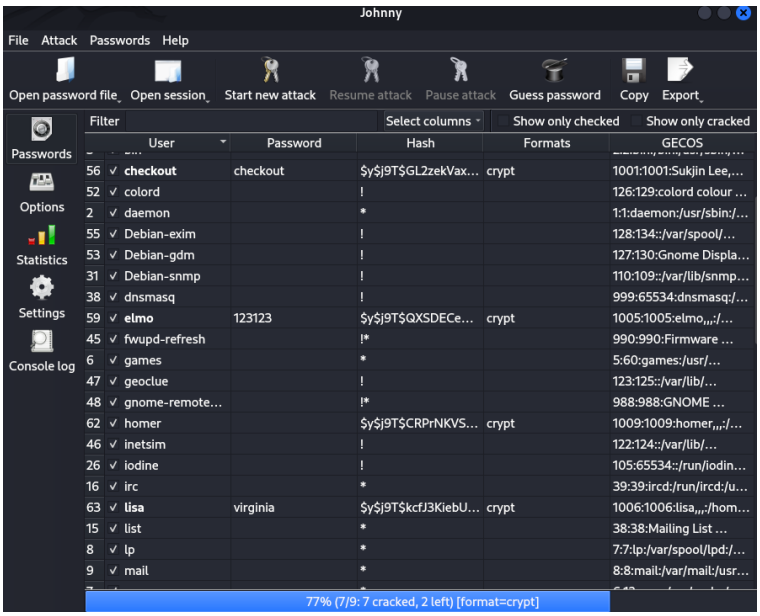


7) Click "Open password file (PASSWD format)". Navigate and find mypasswd file. Click "Open".



8) Click "Options" and select the current hash format: crypt.

9) Click "Start new attack" and then check Passwords. Show the result as shown below:



10) See what happened. Did it successfully crack passwords?
Evan after installing Johnny and updating Linux to make sure everything was up to date the machine said that the command “johnny” was not recognized. So I was not able to complete anything past step 6.

Provide answers to these questions (in this document. Please use a different color for your responses)

1. In the lab, where and how was John the Ripper used? Describe briefly how it works.
   John the Ripper uses the document of potential passwords (john.txt) and attempted them as keywords against the users I had created within the system until it found the passwords. It was used when the command Johnny is called and then the resultant application is used through its own window.
2. How did you get the shadow file from the machine in the lab?
   I got the shadow file by using the “unshadow” command as a root user in the Linux terminal.

Reflection:

1. What was the most challenging part of this assignment for you?
   Making sure that the text file was in the correct spot to be accessed by the following commands.
2. What was the most enjoyable part of this assignment for you?
   Getting to use the terminal within other applications. Using a computer in such a manual way has always been enjoyable for me.
3. What did you learn through this assignment?
   How sensitive information within a system can be accessed and semi “abused” though commands and applications.

Grading Rubric:

	MaxPoints
Screenshots from Lab	4
Answers to questions	4
Spelling/Grammar	2
Total	10

```
crawfodz@ad.jmu.edu@L2020AIO25: ~
elmo:$y$j9T$W40m4DNUs2GhGJH5q1vsn1$cR0GwyOpusVzzqYM4kEbeZPXVOsK5WdEmmW6kV.bkQ4:2
0340:0:99999:7:::
homer:$y$j9T$Rl3hGLuoEjYXEyG8GiRBE0$ccGs5ZDcAxMxXe2rzgmmCin/IIleTNm7RxEM2lg8db2:
20340:0:99999:7:::
crawfodz@ad.jmu.edu@L2020AIO25:~$ john -help
Created directory: /home/crawfodz/.john
Unknown option: "-help"
crawfodz@ad.jmu.edu@L2020AIO25:~$ sudo unshadow /etc/passwd /etc/shadow > mypass
wd
crawfodz@ad.jmu.edu@L2020AIO25:~$ sudo john --format=crypt -wordlist=/home/crawf
odz/Documents/john.txt mypasswd
Loaded 8 password hashes with 8 different salts (crypt, generic crypt(3) [?/64])
Will run 32 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
123123           (elmo)
sanjose          (oscar)
acapulco         (homer)
virginia         (lisa)
4g 0:00:01:02 100% 0.06449g/s 50.14p/s 346.0c/s 346.0C/s protel..zhongguo
Use the "--show" option to display all of the cracked passwords reliably
Session completed
crawfodz@ad.jmu.edu@L2020AIO25:~$ johnny
johnny: command not found
crawfodz@ad.jmu.edu@L2020AIO25:~$
```

```
crawfodz@ad.jmu.edu@L2020AIO25: ~
lisa:$y$j9T$qCHzy/49yXaZKWsKoqIYy1$bS2Lx3rqxkfI.5xx48SV.eUIjY9TFo0nncaweKEtEyC:2
0340:0:99999:7:::
elmo:$y$j9T$W40m4DNUs2GhGJH5q1vsn1$cR0GwyOpusVzzqYM4kEbeZPXVOsK5WdEmmW6kV.bkQ4:2
0340:0:99999:7:::
homer:$y$j9T$Rl3hGLuoEjYXEyG8GiRBE0$ccGs5ZDcAxMxXe2rzgmmCin/IIleTNm7RxEM2lg8db2:
20340:0:99999:7:::
crawfodz@ad.jmu.edu@L2020AIO25:~$ john -help
Created directory: /home/crawfodz/.john
Unknown option: "-help"
crawfodz@ad.jmu.edu@L2020AIO25:~$ sudo unshadow /etc/passwd /etc/shadow > mypass
wd
crawfodz@ad.jmu.edu@L2020AIO25:~$ sudo john --format=crypt -wordlist=/home/crawf
odz/Documents/john.txt mypasswd
Loaded 8 password hashes with 8 different salts (crypt, generic crypt(3) [?/64])
Will run 32 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
123123           (elmo)
sanjose          (oscar)
acapulco         (homer)
virginia         (lisa)
4g 0:00:01:02 100% 0.06449g/s 50.14p/s 346.0c/s 346.0C/s protel..zhongguo
Use the "--show" option to display all of the cracked passwords reliably
Session completed
crawfodz@ad.jmu.edu@L2020AIO25:~$
```

```
crawfodz@ad.jmu.edu@L2020AIO25: ~
ary word
Retype new password:
passwd: password updated successfully
crawfodz@ad.jmu.edu@L2020AIO25:~$ sudo tail -n -4 /etc/shadow
oscar:$y$j9T$Okdlp58KDXnh5REsdRF.A1$Xr4WlVN9rpOqzXVoqnp6ppVzYQzmnkNn3Q9/JXbLt3A:
20340:0:99999:7:::
lisa:$y$j9T$qCHzy/49yXaZKWsKoqIYy1$bS2Lx3rqxkfI.5xx48SV.eUIjY9TFo0nncaweKEtEyC:2
0340:0:99999:7:::
elmo:$y$j9T$W40m4DNUs2GhGJH5q1vsn1$cR0GwyOpusVzzqYM4kEbeZPXVOsK5WdEmmW6kV.bkQ4:2
0340:0:99999:7:::
homer:$y$j9T$Rl3hGLuoEjYXEyG8GiRBE0$ccGs5ZDcAxMxXe2rzgmmCin/IIleTNm7RxEM2lg8db2:
20340:0:99999:7:::
crawfodz@ad.jmu.edu@L2020AIO25:~$ john -help
Created directory: /home/crawfodz/.john
Unknown option: "-help"
crawfodz@ad.jmu.edu@L2020AIO25:~$ sudo unshadow /etc/passwd /etc/shadow > mypass
wd
crawfodz@ad.jmu.edu@L2020AIO25:~$ sudo john --format=crypt -wordlist=/home/crawf
odz/Documents/john.txt mypasswd
Loaded 8 password hashes with 8 different salts (crypt, generic crypt(3) [?/64])
Will run 32 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
123123           (elmo)
```

```
crawfodz@ad.jmu.edu@L2020AIO25: ~
crawfodz@ad.jmu.edu@L2020AIO25:~$ groupadd juniors
groupadd: group 'juniors' already exists
crawfodz@ad.jmu.edu@L2020AIO25:~$ groupadd seniors
groupadd: group 'seniors' already exists
crawfodz@ad.jmu.edu@L2020AIO25:~$ cat /etc/group | grep "seniors"
seniors:x:1005:
crawfodz@ad.jmu.edu@L2020AIO25:~$ cat /etc/group | grep "juniors"
juniors:x:1004:
crawfodz@ad.jmu.edu@L2020AIO25:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
```

```
crawfodz@ad.jmu.edu@L2020AIO25: ~
Setting up sssd-tools (2.9.4-1.1ubuntu6.3) ...
Setting up sssd-ipa (2.9.4-1.1ubuntu6.3) ...
Setting up sssd (2.9.4-1.1ubuntu6.3) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libglib2.0-0t64:i386 (2.80.0-6ubuntu3.4) ...
Processing triggers for libglib2.0-0t64:amd64 (2.80.0-6ubuntu3.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for shared-mime-info (2.4-4) ...
Setting up libgtk-4-1:amd64 (4.14.5+ds-0ubuntu0.5) ...
Setting up libgtk-4-bin (4.14.5+ds-0ubuntu0.5) ...
Setting up gir1.2-gtk-4.0:amd64 (4.14.5+ds-0ubuntu0.5) ...
Setting up tecla (46.0-1ubuntu0.1) ...
Setting up libgtk-4-media-gstreamer (4.14.5+ds-0ubuntu0.5) ...
Setting up simple-scan (46.0-0ubuntu2.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
crawfodz@ad.jmu.edu@L2020AIO25:~$ johnny
johnny: command not found
crawfodz@ad.jmu.edu@L2020AIO25:~$
```